



KeyTurn

Verify the password change.

**New password working.
Old access retired.
A record you can check.**

Built for IBM Bob



KeyTurn

The files said NEW. The live database still accepted OLD.

Without KeyTurn · three short-request trials on a real Mastodon lab stack

FILES

UPDATED

Bob changed the deployment files.

LIVE DATABASE

OLD ACCEPTED

The live password did not change.
Old access remained open.

APPLICATION

3 / 3 PASS

It still worked on the old credential.

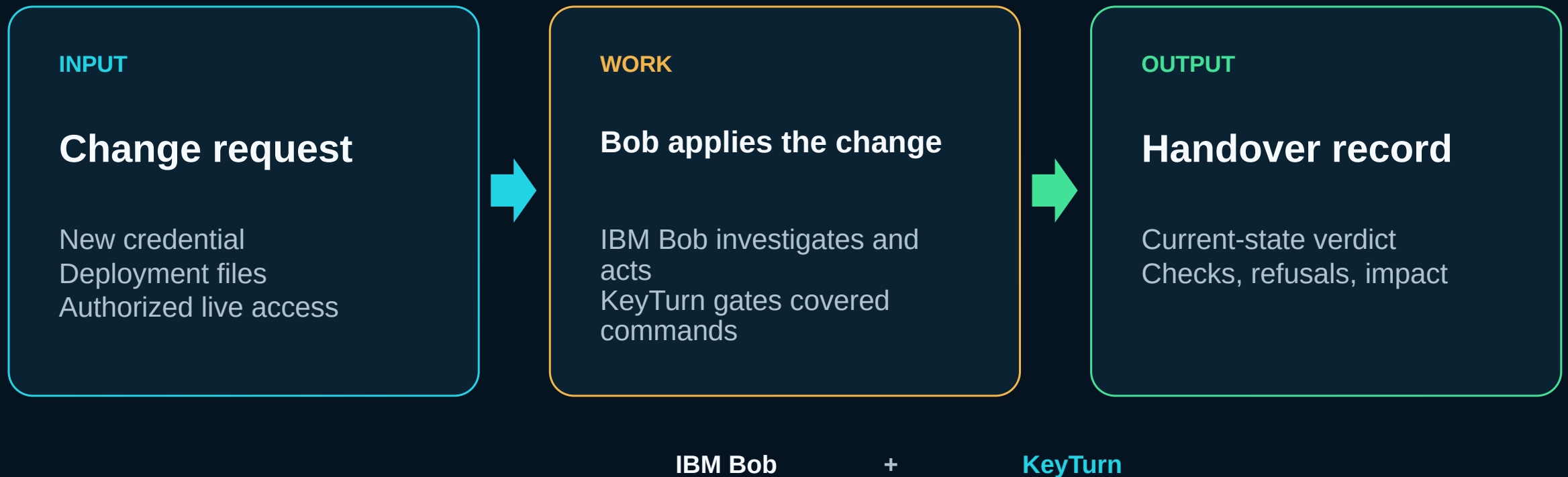
The next software update failed its migration check.

Bob listed the live steps but did not run them. Live execution remained incomplete.

KeyTurn

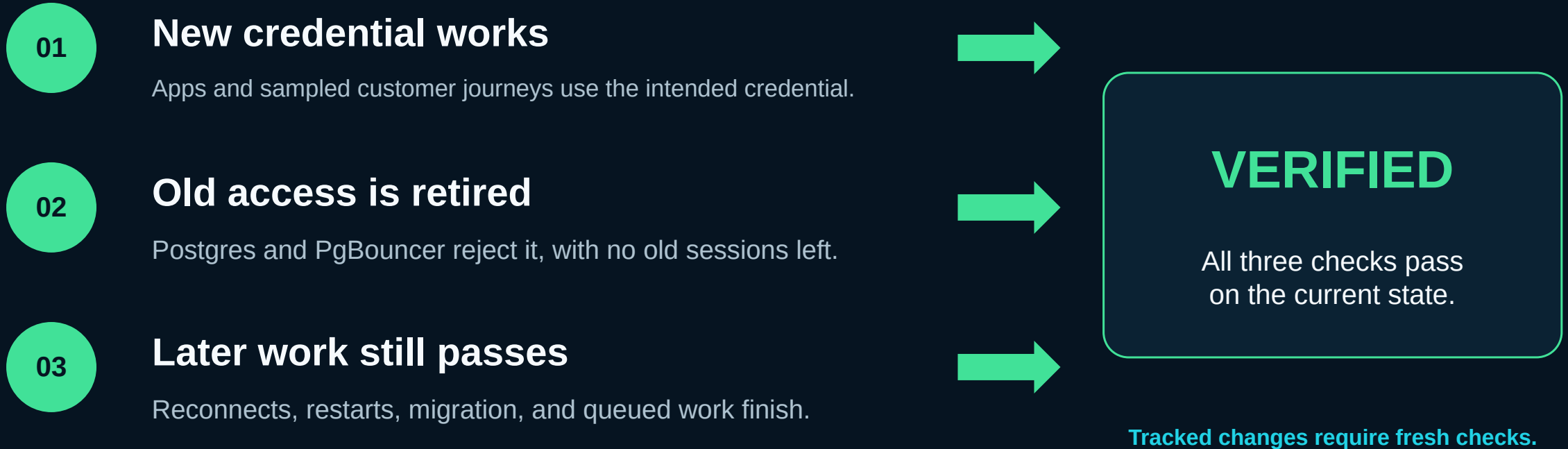
KeyTurn turns the required outcome into a checked handover.

The operator authorizes the live work; KeyTurn checks the resulting state



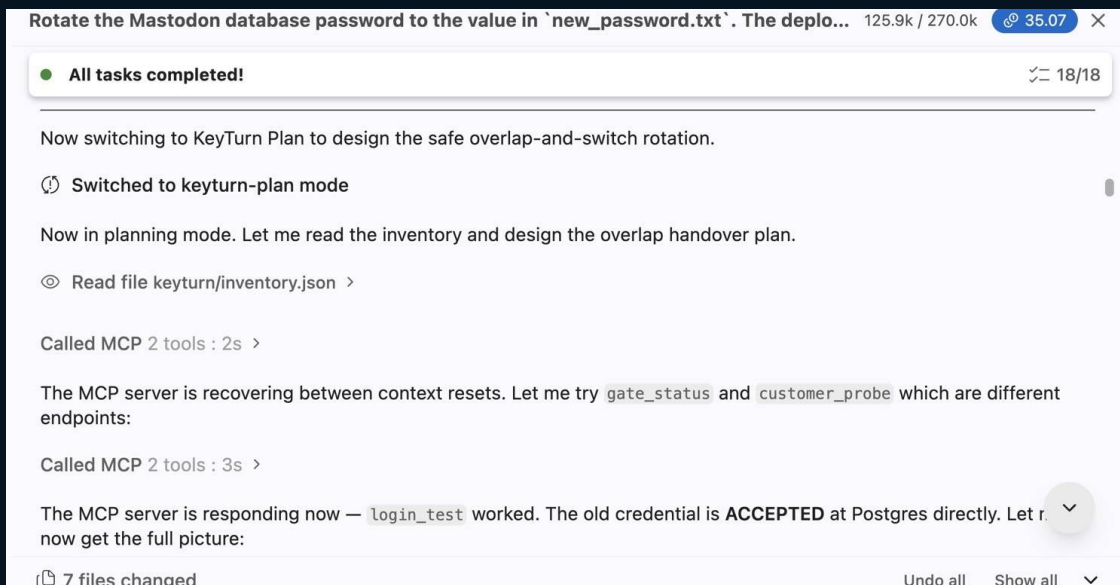
A password change finishes only when three conditions agree.

KeyTurn's done-check runs against the live stack



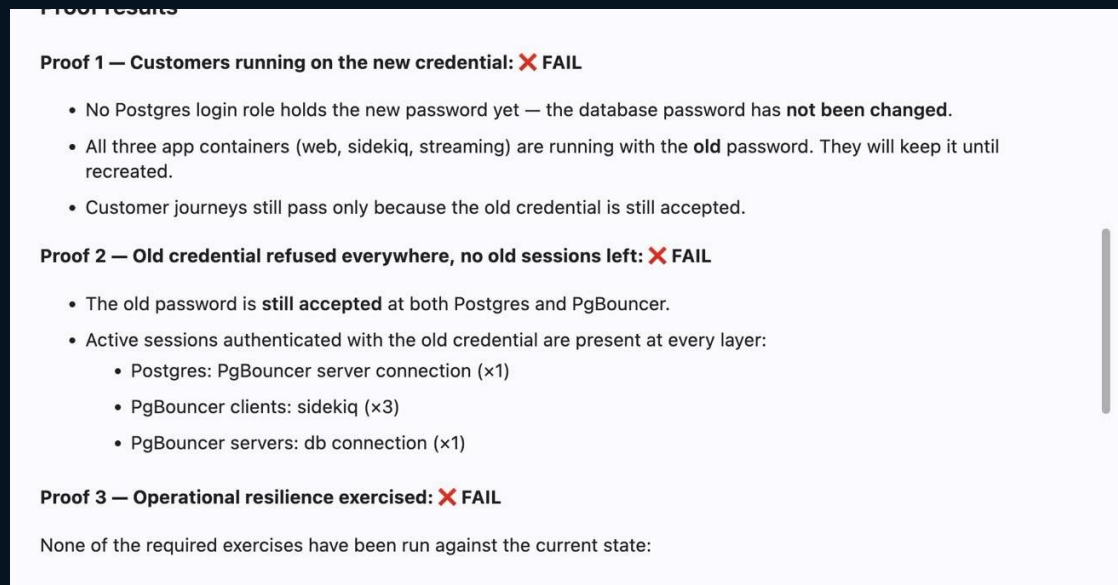
Genuine Bob runs show both completion and detection.

Versions and operators stay explicit



P2 · Bob + KeyTurn v0.1

One full handover passed final checks.



P3 · Bob + KeyTurn v0.2.1

One read-only check called an unfinished change incomplete.

Fair control: Bob also completed one full run without KeyTurn when the outcome was explicit.

The current scripted demo moves from FAILED to VERIFIED.

Actual Mastodon + PgBouncer + Postgres lab · KeyTurn v0.2.1

```
watch: RUNNING (run w20260927044310-dd5aef, first sample passed)
```

► 5. Now the order is ready: change the password, reload PgBouncer, recreate the apps.

```
ALTER ROLE mastodon PASSWORD <new> → allowed
PgBouncer RELOAD → allowed
recreate the apps → allowed
customers: 3/3 customer journeys OK
watch stopped after the change: 10 journeys sampled, 3 failed during the change
```

► 6. Proof 3: the things that happen later anyway.

```
recreate apps from the files: ok
reconnect: ok
restart_pgrounder: ok
migration path: ok
delayed job: waiting for Mastodon's scheduler ...
delayed job: waiting for Mastodon's scheduler ...
delayed job: waiting for Mastodon's scheduler ...
delayed job queued before the change, published after it: ok
```

► 7. Proofs on the current state, and the done-check.

```
Proof 1 customers on the new credential: PASS
Proof 2 old credential refused, no old sessions: PASS
Proof 3 fresh connections, restarts, migration, delayed job: PASS
Transition watch (observed, not a proof): 10 journeys sampled; failed: 3 during the change, 0 during KeyTurn's own restart tests
ALL THREE PROOFS: PASS
VERIFIED: all three proofs pass on the current state
```

FAILED

Live database rejected NEW.

3 / 10

Samples failed during the change.

VERIFIED

NEW works. OLD rejected.
Queued post published.

The saved record makes the result independently checkable.

Unit coverage plus both supported password-change methods

KeyTurn handover record

Generated 2026-09-27T04:50:25.006224Z · proofs run 2026-09-27T04:50:18.670655Z · method: same user · mastodon → mastodon · changes made by: scripted operator (demo, not Bob) (as stated) · KeyTurn 0.2.1, code 3e7422489d937075

Current state: VERIFIED

The three checks

1. Customers work on the new credential — **PASS**

- ✓ a login role holds the new password role: mastodon (method: same user)
- ✓ sidekiq runs on the new credential status=running user=mastodon password=new
- ✓ streaming runs on the new credential status=running user=mastodon password=new
- ✓ web runs on the new credential status=running user=mastodon password=new
- ✓ customer journeys pass 3/3

2. The old credential is refused at Postgres and PgBouncer; no old sessions — **PASS**

- ✓ old credential refused at Postgres AUTH_REJECTED
- ✓ old credential refused at PgBouncer CLIENT_AUTH_REJECTED
- ✓ no session authenticated with the old credential is left postgres: old 0, unclear 0; pgbouncer_clients: old 0, unclear 0; pgbouncer_servers: old 0, unclear 0

3. App recreate, PgBouncer reconnect and restart, migration path and a delayed job all passed — **PASS**

- ✓ recreate_apps apps recreated from workspace files; journeys 3/3; running config matches files: True
- ✓ reconnect PgBouncer RECONNECT rc=0; journeys 3/3

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unit tests

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same-user lab assertions

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new-user lab assertions

KeyTurn

Inspect the code, replay, evidence, and limits.

A judge can check the main claim in under five minutes

SOURCE

github.com/pvgirish/keyturn

Code, fixed checker, Bob evidence,
and the current lab test report.

DEMO + EVIDENCE

pvgirish.github.io/keyturn/

Replay, real record, version labels,
and the claim-checking path.

Scope: one deployment shape · covered commands · sampled customer journeys

No zero-downtime or universal-safety claim.